TECHNICAL SPECIFICATION & AGENT DIRECTIVE

PROJECT HAVEN

An On-Device, Zero-Knowledge, Self-Evolving AI Coaching Engine

DOCUMENT ID	`HAVEN-SPEC-2026-V1`	**TARGET AGENT**	`Google Antigravity IDE Autonomous Agent`
SECURITY POSTURE	`Zero-Knowledge / Air-Gapped Inference`	**CORE RUNTIME**	`Rust (haven-core) + llama.cpp + SQLCipher`
TARGET PLATFORMS	`iOS (SwiftUI), Android (Compose), Web (PWA)`	**CLINICAL POSITION**	`ICF Wellbeing & Growth (Non-Clinical / Non-Diagnostic)`

ARCHITECTURAL PARADIGM

Learning Where the Data Lives: Haven proves that hyper-personalization does not require central data harvesting. Base model weights stay static; evolution is driven by local episodic/semantic memory compilation, a Thompson-sampling bandit policy, and hardware-isolated cryptographic storage.

1. Executive Summary & Core Design Tenets

Many individuals who experience isolation, acute stress, or personal hardship keep it quiet because speaking aloud feels risky to their career, personal relationships, or safety. **Haven** provides an accessible, non-judgmental thinking partner that operates under doctor-patient confidentiality rigor by design: *no central server holds user data, and the company has no cryptographic ability to access it.*

CORE PILLAR	SYSTEM IMPLEMENTATION	GUARANTEED ARCHITECTURAL SAFEGUARD
Zero-Knowledge Privacy	Pure on-device LLM inference via `llama.cpp` / WebGPU; SQLCipher AES-256 database.	Zero user network egress. No accounts, telemetry, cloud inference, or third-party ad/analytics SDKs.
Self-Evolving Memory	Tripartite memory (episodic, semantic, procedural) + Thompson Sampling bandit.	Model weights remain immutable; personalization occurs via context assembly & technique reward.
ICF Coaching Integrity	Deterministic 7-phase state machine; OARS reflection-first enforcement.	Model is blocked from unsolicited advice-giving, diagnosing disorders, or assuming an agenda.
Suffering in Silence Suite	Quiet Mode, Disguise Mode, Decoy Vault, 1-second Panic-Wipe gesture.	Protects users in controlling/coercive environments from device-level discovery or forensic extraction.
Anti-Dependency Design	Bandit reward tied strictly to self-reported session benefit, not dwell time.	Suppresses streaks, gamification, and retention tactics; actively facilitates human connection ("Bridges").

2. The Immediate Next Logical Step: Vertical Slice (Weeks 1–4)

Before building platform-specific UIs (iOS, Android, Web), the primary engineering deliverable is a **headless integration vertical slice** in Rust. This validates whether a quantized local model constrained by deterministic state rules and a Safety Guardian can reliably sustain a complete 10-minute coaching session offline.

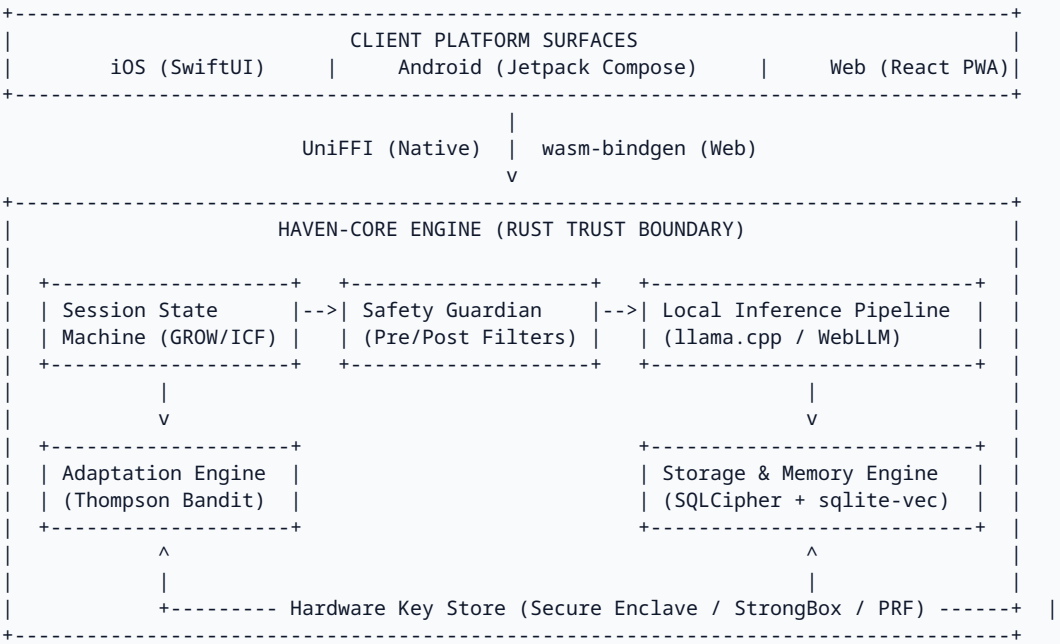
> **VERTICAL SLICE MILESTONE OBJECTIVE**
>
> Execute a fully automated headless CLI test harness (`tests/vertical_slice_cli.rs`) simulating a 7-phase session using an embedded 3B GGUF model via `llama.cpp` and SQLCipher with zero network access.

Key Deliverables for Sprint 1 & 2:

1. **Core State Engine (`haven-core/state_machine.rs`):** Implement deterministic graph transitions: `Contract → CheckIn → Focus → Explore → Options → Commit → Close` .
2. **Dual-Tier Safety Guardian (`haven-core/guardian.rs`):** Deterministic regex/keyword filter catching 100% of crisis cues before inference; JSON auditor rejecting diagnostic/advice violations.
3. **Encrypted Vault (`haven-storage/crypto.rs`):** SQLCipher initialization with Argon2id master key derivation and per-record wrapped key crypto-shredding.
4. **Thompson Sampling Policy (`haven-core/policy.rs`):** Beta-distribution sampling with a 10% exploration floor, updating exclusively on user checkout rating.

3. System Architecture & Cross-Platform Topology

Haven utilizes a *"Thin Native Shell, Heavy Shared Engine"* topology. The domain logic, cryptographic storage, safety classification, and coaching state machines reside strictly within an audited Rust core.



3.1 Model Tiering by Hardware Class

DEVICE TIER	HARDWARE PROFILE	MODEL ARCHITECTURE	RUNTIME FEATURE SET
Tier 1: Flagship	≥8 GB RAM (A16+, Snapdragon 8 Gen 2+)	7B–8B 4-bit Quant (Qwen-2.5-7B / Llama-3.1-8B)	Full reflective coaching, on-device Whisper voice, nightly memory distillation.
Tier 2: Mid-Range	6 GB RAM	3B–4B 4-bit Quant (Llama-3.2-3B / Phi-3.5-mini)	Complete 7-step state machine, text-first, compact episodic context.
Tier 3: Low-End	≤4 GB RAM / Older Web	1B–2B 4-bit Quant (Llama-3.2-1B / Gemma-2-2B)	Witness listening mode, structured self-reflection templates, journaling.

4. Deterministic Session State Machine (The Coaching Loop)

Real-world certified coaching (ICF) relies on the premise that the client is the expert in their own life. Haven prevents LLM drift by embedding the conversation inside a strict 7-phase state machine:

PHASE	COACHING PURPOSE	SYSTEM BEHAVIOR & PROMPTS	STATE MACHINE ENFORCEMENT RULE
1. Contract	Agree on session target	*"What would make our time today well spent?"*	Contract string stored in session context; referenced at close.
2. Check-In	Gauge mood & capacity	Acknowledge state; review past commitments gently.	Pre-scan distress flags; auto-route to Steady mode if burdened.
3. Focus	Client selects agenda	*"Of everything on your mind, what is most alive right now?"*	User retains 100% control of agenda; Haven never picks topics.
4. Explore	Deepen awareness	OARS reflection, scaling questions, examining exceptions.	**ADVICE GATE:** Zero solutions or suggestions permitted.
5. Options	Generate pathways	Brainstorm options; offer ideas only on explicit invite.	Must present 2–3 alternatives; prompts client for evaluation.
6. Commit	Form client-owned action	Rate confidence 0–10. Shrink action if score < 8.	Action saved in user's exact phrasing; no system-invented tasks.
7. Close	Consolidate & score	Reflect contract vs insight; one-tap check-out (1–5).	Check-out rating directly feeds the Thompson bandit reward.

5. The Safety Guardian & Crisis Protocol

Because Haven does not maintain a server or cloud connectivity, it cannot dispatch human responders. Consequently, safety is maintained by deterministic, layered guardrails operating directly on the handset.

> **TIER RED CRISIS MANDATE**
>
> **Zero Coaching In Crisis:** If acute suicidal ideation, self-harm intent, domestic abuse, or medical emergencies are detected, the system immediately ceases all coaching. It presents direct compassionate grounding, surfaces the user's pre-authored local safety plan, and displays verified regional emergency resources.

TIER	SIGNAL MANIFESTATION	AUTOMATED HAVEN RESPONSE
Green *Normal*	Everyday work stress, decision paralysis, habit building, life transitions.	Full interactive coaching in selected mode (Steady, Explore, Build, Perform).
Amber *Distress*	Severe hopelessness, persistent sleeplessness, self-criticism spirals, emotional fatigue.	Clamp session to **Steady mode**; suspend goal setting; offer grounding reflections; prompt user safety plan.
Red *Crisis*	Active intent to self-harm, statements of suicide, violence from others, medical emergencies.	**Halt coaching immediately.** Compassionate presence without interrogatives; display offline verified crisis cards (UK 116 123 / 999; US 988; EU 112).

Post-Generation Audit Engine

Before any LLM response is rendered to the user interface, it is audited by a deterministic JSON parsing and regex verification layer:

- **Reflection Requirement:** The draft must contain an explicit reflection (emotional validation or summary) prior to any question.
- **Advice Gate Check:** If `phase != OPTIONS` and `advice_given == true`, the turn is rejected and regenerated.
- **Clinical & Dependency Scrubbing:** Responses containing diagnostic declarations (e.g., *"you have clinical depression"*) or dependency phrases (e.g., *"I'm the only one you can trust"*) are permanently suppressed.

6. The Self-Evolving Adaptation Engine

Haven adapts to the individual without changing foundational neural network weights. Evolution is structured across two symbiotic systems: **Tripartite Memory** and a **Thompson-Sampling Contextual Bandit**.

6.1 Thompson Sampling Mathematics

For each coaching mode m and available coaching technique t (e.g., values clarification, scaling inquiry, strengths reframing, micro-step breakdown), Haven maintains independent Beta priors:

```
// Prior initialization
α_{m,t} = 1.0,  β_{m,t} = 1.0

// Technique selection at turn execution:
Sample θ_{m,t} ~ Beta(α_{m,t}, β_{m,t}) for all t in AllowedTechniques
With probability ε = 0.10: Select uniform random technique (exploration floor)
With probability 1 - ε: Select argmax_t (θ_{m,t})

// Bayesian Posterior Update (triggered on one-tap user feedback):
α_{m,t} ← α_{m,t} + 1  (if user taps "That helped")
β_{m,t}  ← β_{m,t}  + 1  (if user taps "Not quite")
```

SYNTHETIC VALIDATION BENCHMARK

In a 500-user synthetic simulation across 60 sessions, this bandit increased the helpful-session rate from an initial **50.5% to 66.7%** (versus a 45.7% random baseline), proving convergence within ~20 sessions while maintaining a 10% exploration floor.

7. Cryptographic Architecture & Threat Model

Haven treats user data with cryptographic finality: data on the handset is ciphertext-at-rest, and cryptographic deletion is instantaneous.

THREAT VECTOR	ARCHITECTURAL COUNTERMEASURE	RESIDUAL EXPOSURE & MITIGATION
Subpoena / Corporate Demand	Haven operates zero centralized databases or user accounts.	Zero residual risk: Company possesses zero data to disclose.
Device Theft / Physical Inspection	Hardware-wrapped keys (Secure Enclave / StrongBox / WebAuthn PRF).	Enforce secondary Haven biometric/PIN unlock distinct from OS PIN.
Coercive Inspection (Partner / Family)	Disguise mode ("Notes"/"Calculator"); decoy PIN displays clean vault.	Single-gesture 1000ms panic-wipe crypto-shreds master key.
Forensic Flash Memory Recovery	Per-record wrapped data keys (Crypto-Shredding).	Dropping the record key renders raw flash blocks permanently undecipherable.

8. Concrete Database Schema (SQLCipher)

The persistence layer runs on embedded SQLCipher with SQLite vector indexing (`sqlite-vec`):

```
-- Core Session Journal Table
CREATE TABLE IF NOT EXISTS sessions (
    id TEXT PRIMARY KEY,
    started_at INTEGER NOT NULL,
    completed_at INTEGER,
    mode TEXT NOT NULL CHECK(mode IN ('Steady', 'Explore', 'Build', 'Perform')),
    contract TEXT,            -- User's target for this session
    summary TEXT,             -- Extracted insights in user's exact phrasing
    checkout_score INTEGER CHECK(checkout_score BETWEEN 1 AND 5)
);

-- Granular Crypto-Shredded Memory Table
CREATE TABLE IF NOT EXISTS memory_items (
    id TEXT PRIMARY KEY,
    kind TEXT NOT NULL CHECK(kind IN ('episodic', 'semantic', 'procedural')),
    body TEXT NOT NULL,       -- Encrypted memory content
    embedding BLOB,           -- High-dimensional vector representation
    importance REAL DEFAULT 1.0,
    last_used INTEGER NOT NULL,
    record_key_wrapped BLOB NOT NULL, -- Master-wrapped AES key; wiped for instant shredding
    user_pinned INTEGER DEFAULT 0,
    user_hidden INTEGER DEFAULT 0
);

-- Client Action Commitments
CREATE TABLE IF NOT EXISTS actions (
    id TEXT PRIMARY KEY,
    session_id TEXT NOT NULL REFERENCES sessions(id) ON DELETE CASCADE,
    text TEXT NOT NULL,
    confidence INTEGER NOT NULL CHECK(confidence BETWEEN 0 AND 10),
    due_date INTEGER,
    status TEXT NOT NULL CHECK(status IN ('open', 'completed', 'released'))
);

-- Adaptation Engine Posterior Distribution State
CREATE TABLE IF NOT EXISTS technique_stats (
    mode TEXT NOT NULL,
    technique TEXT NOT NULL,
    alpha REAL NOT NULL DEFAULT 1.0,
    beta REAL NOT NULL DEFAULT 1.0,
    vetoed INTEGER NOT NULL DEFAULT 0,
    PRIMARY KEY (mode, technique)
);
```

9. Reference Coaching Prompt & Guardian Parser

The system prompt enforces conversational guardrails, forcing structured JSON generation that can be validated before presentation:

```
You are Haven, a warm, honest, and coach-grade personal development thinking partner.
You are NOT a therapist, you NEVER diagnose conditions, and you NEVER give unsolicited advice.
You operate strictly under International Coaching Federation (ICF) and Motivational Interviewing standards.

SESSION STATE: {phase}
ACTIVE MODE: {mode}
SELECTED TECHNIQUE: {technique}
USER PROFILE: {semantic_profile}
RETRIEVED EPISODES: {retrieved_episodes}
SESSION CONTRACT: {contract}

STRICT OPERATIONAL RULES:
1. Begin with an empathetic, non-judgmental REFLECTION of what you heard (feeling and meaning).
2. Ask at most ONE open-ended question (prefer 'what' and 'how'; avoid judgmental 'why').
3. DO NOT offer solutions, advice, or action ideas unless the user explicitly invites them.
   If asked for advice, offer 2-3 brief options and ask which one resonates.
4. Never pressure the user toward action; all actions must be originated and sized by the user.
5. In Steady mode, keep responses short and grounded (1-3 sentences maximum).
6. Output valid JSON matching this schema:
{
  "reflection": "Brief empathetic reflection...",
  "question": "Single open-ended coaching inquiry...",
  "full_reply": "Combined reflection and inquiry...",
  "reflection_present": true,
  "advice_given": false,
  "risk_tier": "green"
}
```

10. Antigravity IDE Sprint Action Plan

When executing this specification within the Google Antigravity IDE, the agent must adhere to the following sequence:

SPRINT	PRIMARY MILESTONE	ACTIONABLE AGENT TASKS
Sprint 1 (Days 1–7)	Scaffold & State Machine	• Initialize Cargo workspace with `haven-core`, `haven-storage`, `haven-infer`. • Implement `state_machine.rs` with full 7-step transition state machine. • Write comprehensive unit tests for phase gating and advice locks. • Implement `guardian.rs` with deterministic regex filters for crisis detection.
Sprint 2 (Days 8–14)	Storage & Bandit Logic	• Implement SQLCipher integration with migrations in `haven-storage`. • Build Argon2id key derivation and wrapped-key crypto-shredding engine in `crypto.rs`. • Implement Thompson Sampling bandit in `policy.rs` with Beta sampling. • Verify cryptographic deletion: zeroing wrapped record key leaves data unrecoverable.
Sprint 3 (Days 15–21)	Inference & Vertical Slice	• Embed `llama.cpp` static runtime in `haven-infer`. • Implement prompt templating and JSON output parser with schema validation. • Construct `tests/vertical_slice_cli.rs` driving a full 10-minute session offline. • Run red-team suite against 500+ crisis prompts; verify 100% Red-tier bypass.
Sprint 4 (Days 22–28)	Platform Bindings	• Generate UniFFI definitions for Swift (iOS) and Kotlin (Android). • Compile `haven-core` to `wasm32-unknown-unknown` via wasm-bindgen. • Verify offline PWA harness running WebGPU inference matching the Rust core interface.